



Society of Petroleum Engineers

SPE-229071-MS

An Effective Way to Generate Hydrogen and Nano Material from Hydrogen Sulfide

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Copyright 2025, Society of Petroleum Engineers DOI [10.2118/229071-MS](https://doi.org/10.2118/229071-MS)

This paper was prepared for presentation at the ADIPEC held in Abu Dhabi, UAE, 3 – 6 November 2025.

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Abstract

Hydrogen sulfide (H_2S) is a toxic byproduct that requires treatment and conversion to useful chemicals, and thus production of hydrogen (H_2) using H_2S is beneficial and valuable in industry. Various technical ways have been developed focusing on splitting H_2S into H_2 and sulfur (S). In this work we propose a novel electrochemical technology for generating high purity H_2 and high value nano zinc sulfide (ZnS) from H_2S .

The developed system consists of two integrated units: an electrolysis unit for H_2 generation and a production unit for synthesis of sulfide derives from H_2S . Zinc (Zn) plate serves as soluble anode to generate zinc ions through anodic dissolution during electrolysis, while high purity H_2 and hydroxyl ions are generated at cathode. The generated zinc hydroxide, chelated with specialized electrolytes, is transported to the production unit where it reacts with H_2S under controlled conditions to produce nano zinc sulfide (ZnS) and regenerate the electrolytes, creating a recyclable electrolytes system for H_2 generation and ZnS production.

The experimental results demonstrate the efficiency and reliability of the developed electrochemical system. This small device can produce 0.46 L/hr H_2 with energy consumption of approximately 2.55~3.03 Kwh/m³. The alkaline zinc ion solution reacts with simulated H_2S to generate ZnS precipitation. X-ray diffraction (XRD) analysis and scanning electron microscopy (SEM) images prove the achieved ZnS is a high purity nanometer-size product with cluster size around 50~200 nm. The electrolytes used in the system can be recycled to consistently produce high purity H_2 and nanosized ZnS with the same cluster size, proving the efficacy of the developed recyclable system.

This technology represents a new approach with advancements to convert hazardous H_2S into high purity H_2 with low energy consumption and nano zinc sulfide (ZnS) as a high-value product. It offers a sustainable pathway to provide low-carbon energy and advanced materials with zero waste, enhancing resource utilization efficiency and reducing the environmental impact of H_2S .

Introduction

Hydrogen sulfide (H_2S) as a highly toxic and corrosive by-product is produced with significant quantities in oil and gas production and petroleum refining processes (Beauchamp, Bus et al. 1984, Cox, Clarke et al. 1998, Thimm 2008, Reverberi, Klemeš et al. 2016). Recognized by its distinctive rotten eggs odor, H_2S poses severe health and safety risks, with the potential to result in adverse health impacts and fatal outcome even under extremely low concentrations and short exposure durations (Beauchamp, Bus et al. 1984, Chan, Lock et al. 2022). Therefore, effective containment, treatment and disposal of this hazardous gas is critical for operation and management across the energy sector. In upstream operations, chemicals such as amines are typically added as the most common mitigation strategy to capture H_2S from produced gas streams. These H_2S -laden compounds are then transported to downstream facilities for further purification of H_2S and regeneration of the chemical agents for recycling usage (Obakore W. Agbroko 2017).

In the refining industry, the Claus process represents an industry-standard technology for H_2S treatment. This mature solution involves thermocatalytic conversion of H_2S into element sulfur (S) and water via a sulfur dioxide (SO_2) intermediate at high temperatures (Sassi and Gupta 2008, Zhang, Tang et al. 2015). Despite the Claus process being highly efficient in conversion of H_2S gas with high sulfur recovery, it is also energy-intensive with significant CO_2 emissions. Moreover, the combustion of residual H_2S and other unconverted impurities requires further treatment of the generated byproduct SO_2 after the process (Chan, Loy et al. 2023). From a resource utilization perspective, this process results in water generation, representing a missed opportunity to utilize hydrogen source to recover valuable energy content.

The drive to reduce carbon intensity with enhancing resource utilization efficiency is accelerating the development of innovative and sustainable alternative technologies for H_2S management. Theoretically, H_2S can be split into H_2 and S, offer benefits for low-carbon energy generation, sulfur production and hazardous chemicals processing. In the past several decades, Various techniques (De Crisci, Moniri et al. 2019) have been developed including thermal catalytic (Reshetenko, Khairulin et al. 2002), thermal chemical (Towler and Lynn 1993), photocatalytic (Dan, Yu et al. 2020), non-thermal plasma (Gutsol, Nunnally et al. 2012, Linga Reddy, Biju et al. 2012), electrochemical (Weaver and Winnick 1987, Mao, Anani et al. 1991, Mizuta, Kondo et al. 1991, Mbah 2008, Huang, Shang et al. 2019), biological methods (Sakurai, Masukawa et al. 2013) and other methods (Zhang, Zhang et al. 2019). While these technologies prove technical feasibility and potential for hydrogen recovery, none of them have yet reached commercial scale deployment due to economic issues and technical challenges. Therefore, there remains significant opportunities for development of cost-effective methods that can transform hazardous H_2S into a valuable resource for low-carbon energy and value-added chemicals such as sulfur and sulfur derivatives.

In this work, a new way for production of H_2 from H_2S was developed by using a designed two-unit system including an electrolytic unit and a production unit. In this system, Zinc (Zn) plate is used as anode to produce zinc ions (Zn^{2+}) through anodic dissolution reaction. High purity H_2 and hydroxyl ions (OH^-) are generated on the cathode. Soluble zinc hydroxide ($\text{Zn(E)}_x(\text{OH})_2$) achieved by the produced ions with the help of developed electrolytes (E) then is introduced into production unit, where it reacts with H_2S gas to produce nano ZnS as value-added product and regenerate electrolytes, creating a recyclable electrolytes system for H_2 generation and ZnS production. Compared with the previous technologies, this new electrochemical way can produce H_2 with high purity and nano zinc sulfide (ZnS) with high value. This work also investigated H_2 production rate and energy consumption for the designed lab-scale electrolysis unit and characterized purity and morphology of produced ZnS products separately. The overall performances with regenerated electrolytes were evaluated to explore feasibility of establishing a recycling system.

In this paper, an innovative electrochemical system was developed to provide a sustainable method for H_2 generation from H_2S , while simultaneously producing nanosized zinc sulfide (ZnS) with high value. The designed system includes an electrolytic module and a production module with a closed-loop configuration.

In the electrolytic unit, zinc (Zn) plate serves as the anode to produce zinc ions (Zn^{2+}) through anodic dissolution reaction, while high purity H_2 and hydroxyl ions (OH^-) are generated on the cathode. These ions interact with a specially formulated electrolyte (E) to form a soluble zinc hydroxide complex ($\text{Zn(E)}_x(\text{OH})_2$). Then, the intermediate is transferred into production unit, where it reacts with H_2S gas to produce nanoscale ZnS product and regenerate electrolytes, creating a recyclable electrolytes system for conversion of H_2S into H_2 and ZnS. Compared to conventional H_2S treatment technologies, this electrochemical approach can offer benefits including generation of high purity H_2 without purification and providing nanomaterials with high value. The system was further evaluated at laboratory scale with investigation of hydrogen production rates, energy consumption and ZnS product characteristics. Additionally, the performance of the system using regenerated electrolytes was assessed to validate the recycling feasibility of the closed-loop operation.

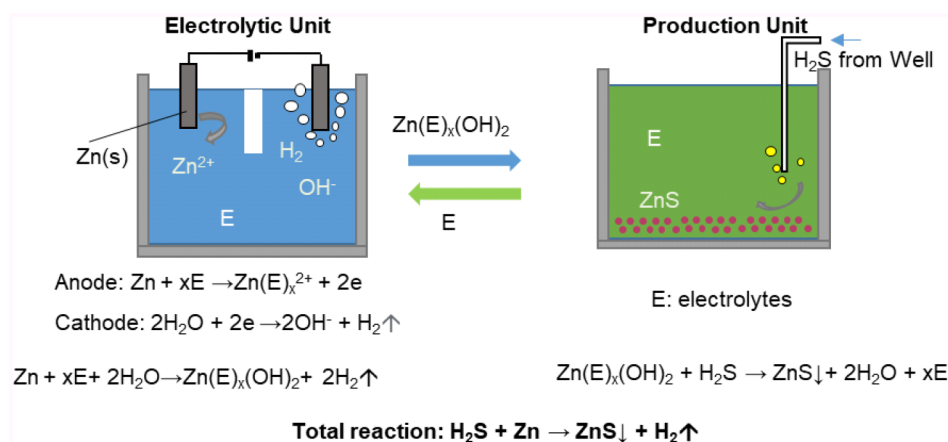


Figure 1—Schematic diagram of H_2 generation from H_2S

Experimental

Materials

Chemicals. Zinc plate ($50\text{mm} \times 50\text{mm} \times 0.25\text{mm}$), titanium plate ($50\text{mm} \times 50\text{mm} \times 0.2\text{mm}$), sodium sulfide (Na_2S) and sulfuric acid (H_2SO_4) purchased from Innochem company are all analytical pure. Water is filtered and deionized in an ultra-pure water system.

Instruments. DC power supply (PDS3030-10s, model: 0~30V, 0~10 A) is purchased from Loitest Company, Suzhou. Titanium plate ($50\text{mm} \times 50\text{mm} \times 0.2\text{mm}$), and 1000 ml electrolysis cell with cap are ordered from Ledon Company, Shanghai. Two automatic dosing pumps (TPG 500 and TPG 600) are purchased from SEKO company, Italy. Hydrogen fuel cell is purchased from Integrity Teaching Instrument Satiation, Taizhou.

Electrolysis System

H_2 generation is conducted in a designed electrochemical reactor. The designed electrochemical reactor includes a direct current (DC) power supply, a 1000 mL electrolysis chamber, and a magnetic stirrer to ensure uniform electrolyte mixing. The electrolyzer equipped with a titanium plate ($50\text{mm} \times 50\text{mm} \times 0.2\text{mm}$) as cathode, and a zinc plate ($50\text{mm} \times 50\text{mm} \times 0.25\text{mm}$) as anode, with a fixed inter-electrode distance of 20 mm (Fig. 2). This electrochemical module is integrated with a feed tank, a product collection tank, and a hydrogen fuel cell that can convert the generated H_2 into electricity to power a connected fan, demonstrating the system's potential for energy recovery and functional application. (Fig. 3).

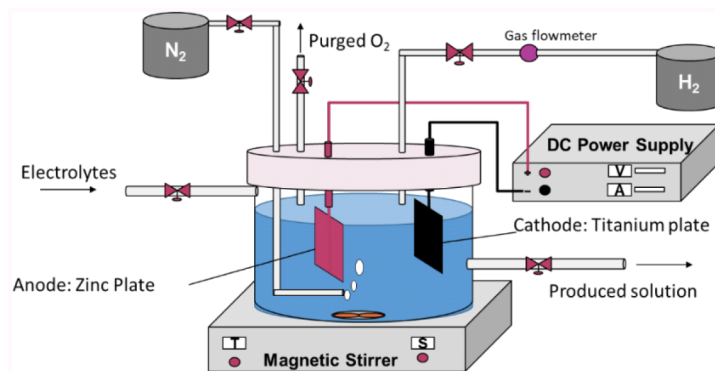


Figure 2—The designed electrolysis cell

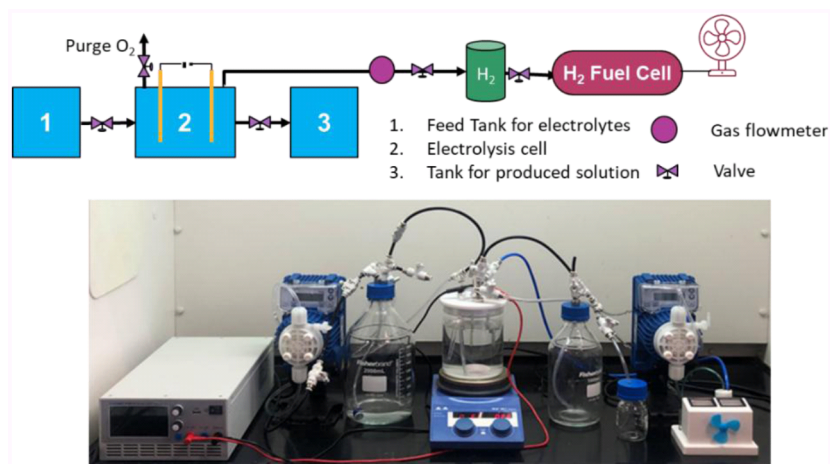


Figure 3—Assembled electrolysis unit for hydrogen generation

H₂ Production Rate and Energy Consumption Measurement

A pre-calculated volume of the prepared electrolyte solution is introduced into the electrolysis chamber from the storage tank. A pre-weighed zinc plate as the soluble anode, and a titanium plate as the cathode are fully immersed into the electrolyte solution. To eliminate dissolved oxygen and residual air within the system, high-purity nitrogen gas is purged through the setup for 30 minutes. Following the purge, a DC power supply is activated and adjusted to the designated voltage to initiate hydrogen generation. Throughout the electrolysis process, voltage and current are monitored and recorded for calculation of energy consumption. Upon completion, the zinc anode is reweighed to determine the mass loss, which is used to quantify the amount of hydrogen produced based.

H₂ production rate (v) is calculated by equation (1):

$$v = \frac{m_i - m_e}{M_{Zn} \times t} \times V_m \quad (1)$$

where m_i is the initial mass of zinc plate (g); m_e is the final mass of zinc plate after an experiment (g); M_{Zn} is the molecular weight of zinc (g/mol); t is the length of time between initial and final readings; V_m is the standard volume of gas (~22.4 L/mol).

Energy consumption (EC) for H₂ production is calculated by equation (2):

$$EC = \frac{V \times A \times t \times M_{Zn}}{(m_i - m_e) \times V_m} = \frac{V \times A}{v} \quad (2)$$

where V is the set voltage for electrolysis (volts); A is the recorded current (amps); v is the H₂ production rate (L/hr).

Synthesis of Zinc Sulfide

The produced zinc hydroxide solution with electrolytes is subsequently transferred into a 1000 ml three-neck flask equipped with magnetic stirring at 25°C. Two constant-pressure dropping funnels are prepared with 1 mol/L Na₂S solution and 1 mol/L H₂SO₄ solution, respectively. These solutions are simultaneously added dropwise into the reaction flask to simulate H₂S gas. The immediate formation of white turbidity in the solution indicates the successful precipitation of zinc sulfide (ZnS). The addition of both reagents is stopped once the solution reaches a neutral pH of 7. The reaction mixture is then continuously stirred for an additional 2 hours at 25 °C to ensure complete conversion. After that, the generated ZnS precipitate is separated by filtration and dried in an oven at 120 °C for 24 hours, yielding a light-yellow powder (Figure 4). The regenerated electrolyte solution is collected and reserved for subsequent electrolysis cycles to demonstrate the closed-loop operation of the system.

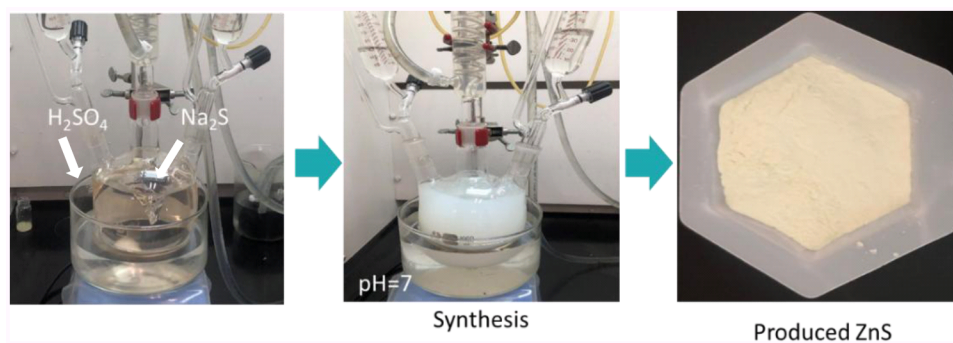


Figure 4—Synthesis of zinc sulfide from stimulated H₂S

X-ray Diffraction Measurement

The crystalline structure of ZnS is characterized using powder XRD measurement. Diffraction patterns are recorded on a Rigakua Smartlab X-ray powder diffractometer via using CuKα irradiation, operating at room temperature in the 2θ range 5-90° with a scan rate of 10°/min.

The interplanar spacing between atoms (*d*) is calculated using Bragg's law:

$$d = \frac{n\lambda}{2\sin\theta}$$

The lattice constant (*a*) is calculated by the following equation:

$$a = d\sqrt{h^2 + k^2 + l^2}$$

The particle size (*D*) has been estimated by using Debye-Scherrer formula:

$$D = \frac{0.9\lambda}{\beta\cos\theta}$$

where λ is electron wavelength of X-ray around 0.1541 nm, *n* is order of interference (in this case *n* is 1), θ is the diffraction angle, *h*, *k*, *l* are the values of plane (*hkl*), β is FWHM (full width at half maximum).

Scanning Electron Microscopy Measurement

The morphology of synthesized ZnS is characterized by SEM. SEM images are acquired from a Hitachi SU8000 SEM.

Results

Electrolysis for H₂ Generation in Fresh Electrolytes

The electrolysis reaction was conducted in a freshly prepared electrolytes solution with adjusting pH of approximately 7 at 25°C. The current setting at 1 ampere (A) was applied to initiate the electrochemical reaction. The electrolysis was stopped once the pH of the solution reaches at 10, indicating sufficient progression of the anodic and cathodic processes. As illustrated in Fig.5, the zinc plate as anode released Zn^{2+} ions through anodic dissolution process, while H_2 bubble were visibly generated at the titanium cathode. The resulted solution remained transparent, indicating good solubility and stability of the formed zinc hydroxide complex $(\text{Zn(E)}_x(\text{OH})_2)$ due to the developed electrolytes formulation. The formulation can effectively chelate Zn^{2+} ions, significantly improving the solubility of zinc hydroxide $(\text{Zn}(\text{OH})_2)$ and preventing precipitation at high pH condition.

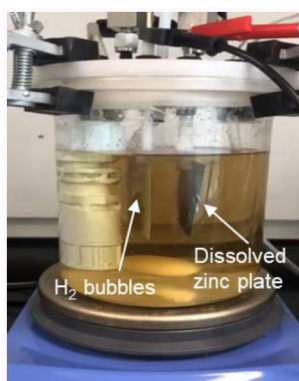


Figure 5—Electrolysis process for H₂ generation

The evolved hydrogen gas was directly transferred into a hydrogen fuel cell, where it was successfully converted into electricity to drive a connected fan (Video 1, Supplementary Material). These results confirm that the developed system can generate high purity of hydrogen exceeding 95% without purification as such purity is required for efficient fuel cell conversion. Fig. 6 presents the hydrogen production performance, displaying an average generation rate of approximately 0.46/L through this small device. The corresponding energy consumption is also calculated to be around 2.55 Kwh/Nm³ of H₂. Compared to conventional water electrolysis, this technology achieves over 46.87% reduction in energy consumption (Fig. 7). Additionally, it demonstrates energy efficiency on par with traditional H₂S electrolysis technologies, even without system-level optimization, highlighting its promising potential for further performance enhancement in hydrogen production rate and energy consumption reduction.

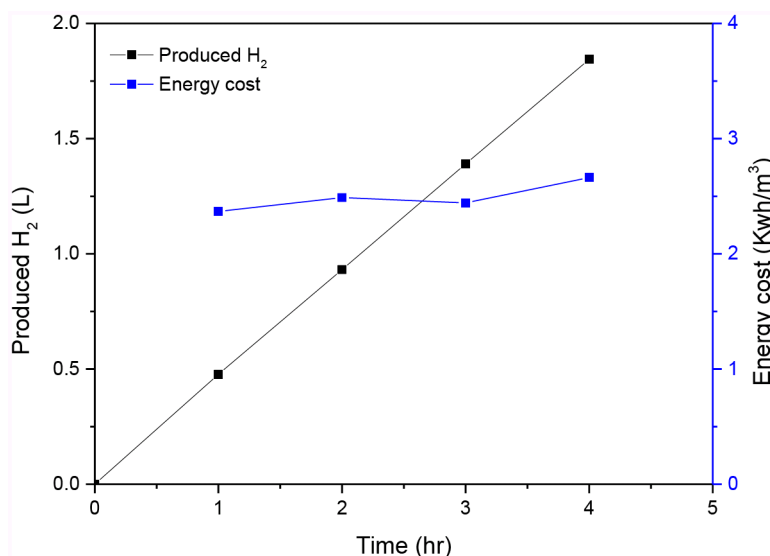


Figure 6—H₂ generation rate and energy cost using fresh electrolytes

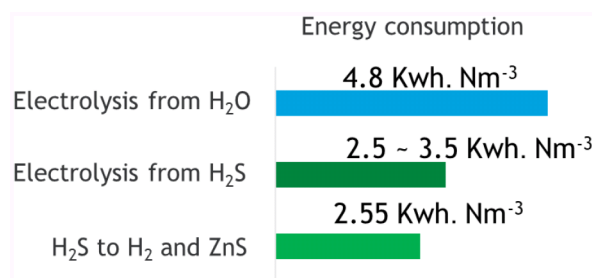


Figure 7—Energy consumption comparison with conventional water electrolysis and H₂S electrolysis

Characterization of Synthesized ZnS from Fresh Electrolytes

To simulate H₂S gas, 1 mol/L Na₂S solution and 1 mol/L H₂SO₄ were simultaneously introduced as dropwise into the generated Zn(E)_x(OH)₂ solution with a controlled rate of 100 ml/hr, which was benefit for nanoscale ZnS synthesis. The resulting precipitations were purified and subsequently characterized by XRD and SEM to evaluate their crystalline structure and morphological features.

XRD Analysis for Synthesized ZnS. The XRD pattern of synthesized product is shown in Fig. 8. The structural parameters of ZnS obtained from XRD measurement are summarized in Table 1. The three prominent diffraction peaks at 28.67°, 47.64°, and 56.61° are observed, which correspond to (111), (220), and (311) planes of the cubic crystalline ZnS, respectively. These reflections are consistent with ZnS standard card (JCPDS, No. 05-0566), confirming the formation of a cubic zinc blende structure. The peaks are noticeably broadened due to the small particle size of the sample, which indicates the successful synthesis of nanosized ZnS. Based on the peak analysis, the average crystallite size for the produced ZnS is around 6.40 nm, demonstrating the successful synthesis of nanostructure material.

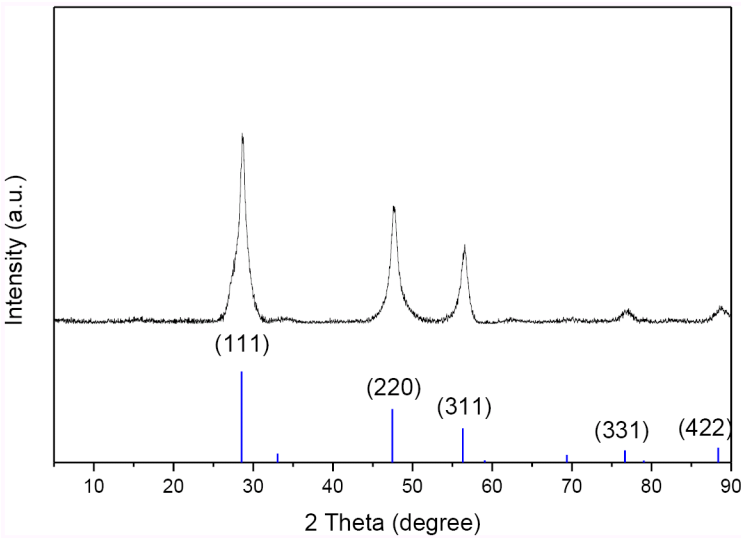


Figure 8—XRD patterns of synthesized ZnS from fresh electrolytes

Table 1—Structural parameters of synthesized ZnS from fresh electrolytes

2θ (degree)	Plane (hkl)	Interplaner Spacing d (Å)	FWHM β (rad)	Lattice Constant a (Å)	Crystallite Size (nm)	
					Size	Average size
28.67	111	3.12	0.023	5.40	6.11	6.40
47.64	220	1.91	0.025	5.39	6.02	
56.61	311	1.62	0.022	5.39	7.07	

SEM Images Analysis for Synthesized ZnS. SEM images in Fig. 9 visually confirm the successful synthesis of nanoscale ZnS using simulated H₂S and generated Zn(E)_x(OH)₂ solution through electrolysis. The achieved nanoparticles appear in cluster formations which exhibit a distinct cauliflower-like morphology, consistent with previously reported structures in the literature on nano ZnS synthesis (Liu, Guo et al. 2014, Park, Park et al. 2014). The observed cluster sizes range around 50~200 nm, further demonstrating the formation of nanostructures.

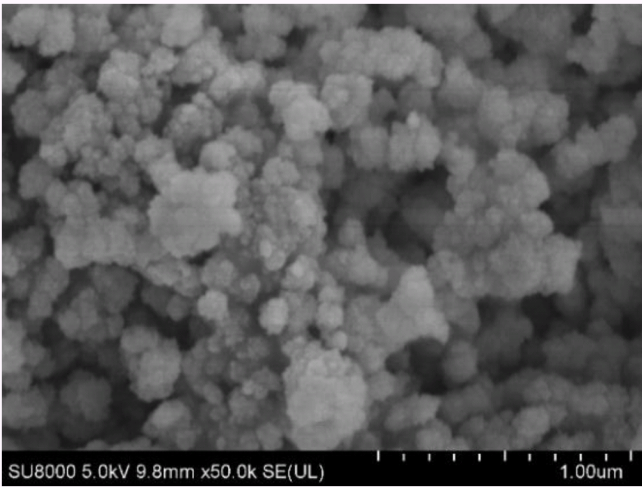


Figure 9—SEM image of synthesized ZnS using fresh electrolytes

H₂ Generation and ZnS Production in Regenerated Electrolytes

The regenerated electrolyte solution was utilized for H₂ generation and ZnS production. In this study, the electrolyte undergoes two regeneration cycles. Electrolysis experiments were performed with a constant current at 1 A and stopped once the solution pH reaches at 10. The resulting solutions from the regenerated electrolytes were subsequently reacted with simulated H₂S gas under the same conditions applied to the fresh electrolyte system (as illustrated in Fig. 10).

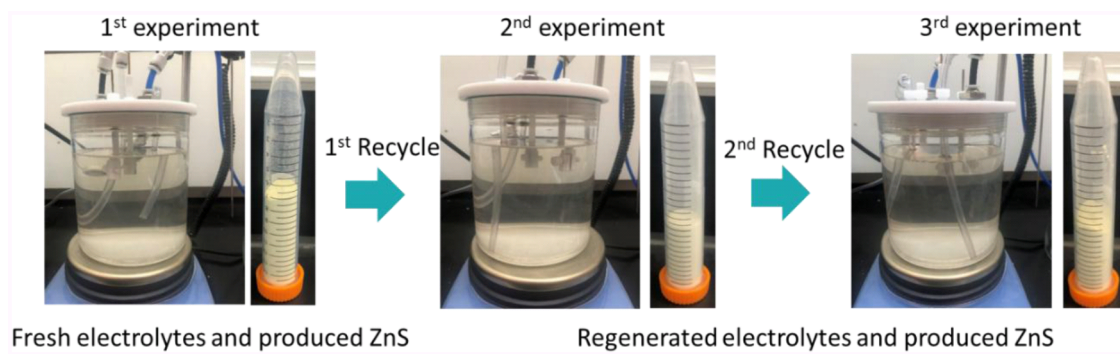


Figure 10—Electrolytes and produced ZnS before and after the recycling experiments

H₂ Production in Regenerated Electrolytes. Fig. 11 presents H₂ generation results and energy consumption in the two recycling experiments. The H₂ producing rate in the first recycling experiment is around 0.44 L/hr with energy consumption around 3.0 Kwh/Nm³ of H₂ and the H₂ producing rate in the second recycling experiment is around 0.46 L/hr with energy consumption around 3.03 Kwh/Nm³ of H₂. The results indicate that the electrolytes can be effectively regenerated for H₂ generation. It is interesting to notice that energy consumption is slightly increased compared to that observed with fresh electrolytes. This may be due to using simulated H₂S gas instead of real H₂S gas. The addition of Na₂S and H₂SO₄ solutions during simulation can dilute the original concentration of the developed electrolytes. Although the introduced Na⁺ and SO₄²⁻ ions may enhance the conductivity of electrolytes solution, potentially mitigating the impact of dilution and contributing to improved electrolysis performance, they cannot fully compensate for the negative effect from dilution, resulting in a net increase in energy consumption.

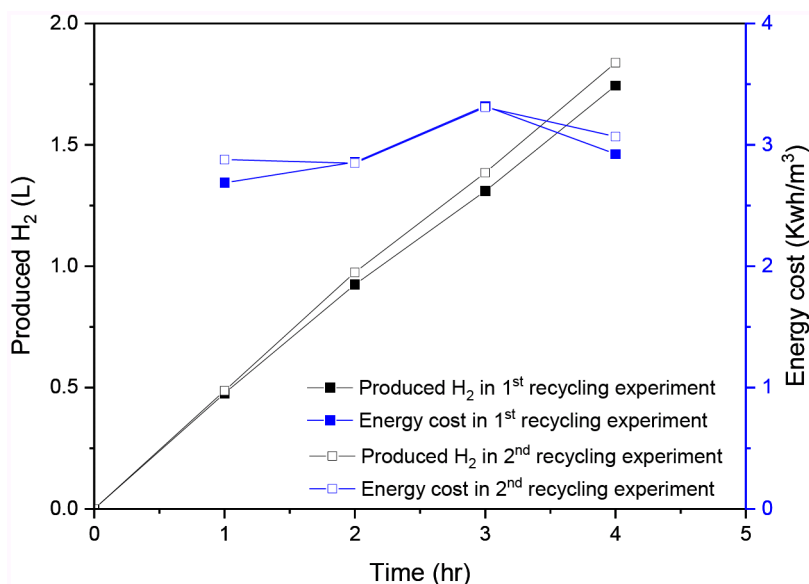


Figure 11—H₂ generation rate and energy cost using regenerated electrolytes

XRD Analysis for Synthesized of ZnS Using Regenerated Electrolytes. Fig. 12 presents the XRD patterns of synthesized products from the first recycling experiment and the second recycling experiment. The structural parameters obtained from XRD measurement are listed in Table 2. The results indicate the synthesized products are cubic crystalline ZnS nanoparticles with sizes around 6.71~6.91 nm. The results prove the electrolytes can be effectively recycled for H₂ generation and nanosized ZnS production with high purity, thereby validating the establishment of a robust and sustainable closed-loop system.

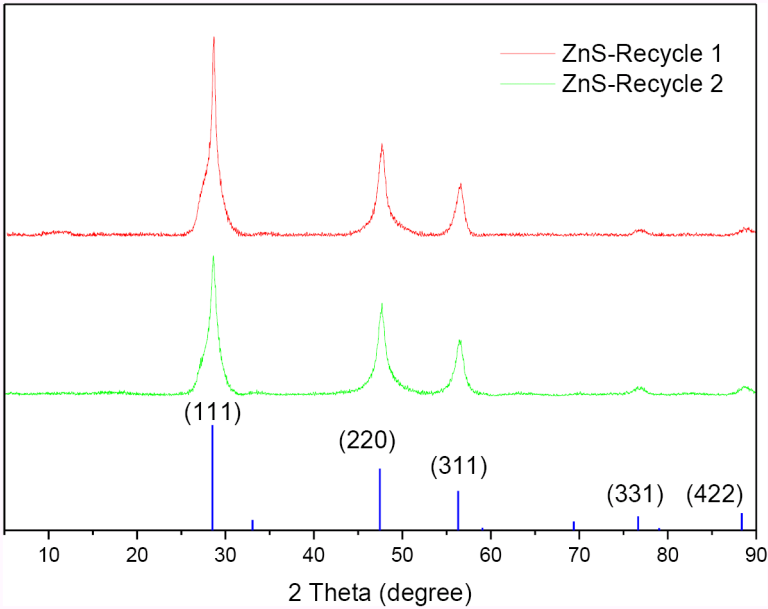


Figure 12—XRD patterns of synthesized ZnS from recycling experiments

Table 2—Structural parameters of synthesized ZnS from recycling experiments

Sample	2θ (degree)	Plane (hkl)	Interplanar Spacing d (Å)	FWHM β (rad)	Lattice Constant a (Å)	Crystallite Size (nm)	
						Size	Average Size
ZnS-Recycle 1	28.69	111	3.11	0.020	5.39	7.11	6.91
	47.65	220	1.91	0.026	5.39	6.12	
	56.68	311	1.62	0.018	5.38	8.49	
ZnS-Recycle 2	28.56	111	3.12	0.024	5.39	5.90	6.71
	47.63	220	1.91	0.020	5.39	7.59	
	56.54	311	1.63	0.023	5.38	6.65	

SEM Images Analysis for Synthesized ZnS. SEM images in Fig. 13 prove that nanosized ZnS has been successfully synthesized by simulated H₂S and Zn(E)_x(OH)₂ produced from regenerated electrolytes. Fig. 13 also reveals cauliflower-like clusters with particle sizes around 50~200 nm. The results are promising as there is no obvious difference between the synthesized products by using regenerated electrolytes and those obtained from fresh electrolytes. This demonstrates nanosized ZnS products can be reliably achieved using regenerated electrolytes.

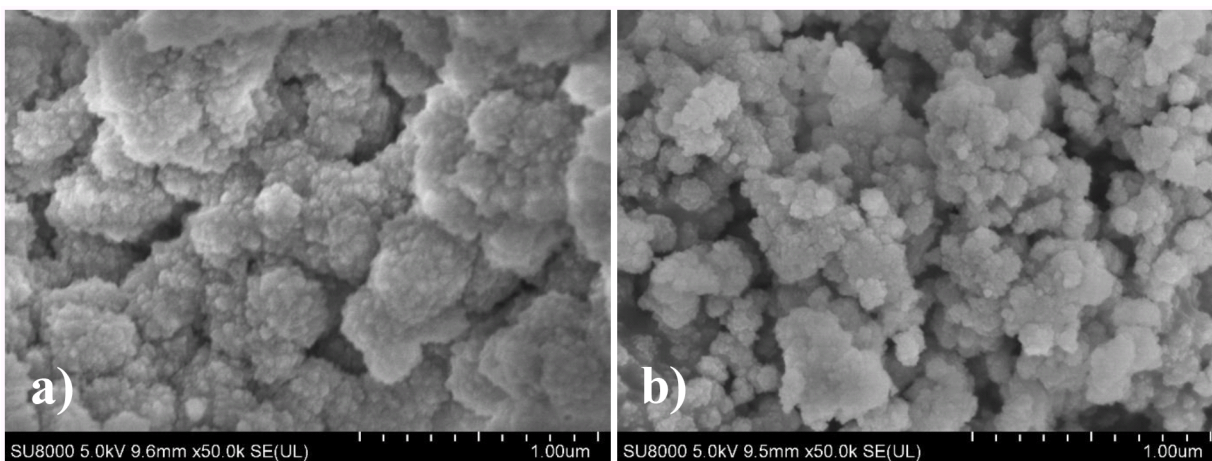


Figure 13—SEM image of synthesized ZnS from: a) First recycling experiment; b) second recycling experiment

Conclusions

1. A novel electrolysis method was proposed for converting H_2S gas into high purity H_2 and nano ZnS with high value. A well-design electrochemical system was successfully established. The developed small electrolytic device can achieve H_2 production rate around 0.46 L/hr with energy consumption around 2.55 Kwh/ Nm^3 of H_2 . This method presents low energy consumption and produces high purity H_2 that can be directly used to generate electricity by H_2 fuel cell.
2. The produced zinc hydroxyl solution stabilized through chelation in the presence of the developed electrolytes exhibits good solubility. It was subsequently reacted with simulated H_2S gas to get nanoscale ZnS while regenerating electrolytes solution. The XRD analysis and SEM images prove the achieved ZnS is a high purity nanosized product with cluster size around 50~200 nm.
3. The regenerated electrolytes were successfully reused for further H_2 generation and ZnS synthesis. In the two recycling experiments, the H_2 production rate remains stable at approximately 0.44~0.46 L/hr with energy consumption around 3.00~3.03 Kwh/ Nm^3 of H_2 , demonstrating the establishment of an effective and sustainable closed-loop system for converting H_2S into H_2 and nanosized ZnS.

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